

Independent Replication — *Adiabatic Quantum State Generation and Statistical Zero Knowledge*

Aharonov & Ta-Shma, arXiv:quant-ph/0301023 (2003)

Independent replication (QC-200 wave, 2026-07-05)

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1 Paper summary

Aharonov and Ta-Shma (2003) introduce the paradigm of *adiabatic quantum state generation*: instead of building unitary circuits, one specifies the target state $|\psi\rangle = \sum_x \sqrt{p(x)} |x\rangle$ (the “coherent encoding” of a classical distribution p) as the *ground state* of a final Hamiltonian H_1 and adiabatically transports an easily-prepared starting ground state (uniform superposition $|+\rangle^{\otimes n}$, ground state of some H_0) into $|\psi\rangle$ via the straight-line path

$$H(s) = (1 - s)H_0 + sH_1, \quad s \in [0, 1].$$

The paper’s main contributions:

- **Theorem 1 (SZK $\Rightarrow$ Q-sampling).** Every promise problem in Statistical Zero Knowledge (SZK) reduces to *coherent quantum sampling* (“Qsampling”) from an efficiently-samplable classical distribution — via a reduction from the SD (Statistical Difference) complete problem for SZK, using the “ $|v\rangle + |w\rangle$ ” Hadamard-test trick of their Claim 1.
- **Theorem 2 (equivalence).** Any state generatable efficiently in the standard circuit model is also efficiently generatable adiabatically, and vice versa.
- **Lemma 1 (sparse-Hamiltonian).** Any row-sparse, row-computable Hamiltonian of polynomial norm is simulatable.
- **Lemma 2 (jagged adiabatic path).** A sequence of Hamiltonians with non-negligible spectral gaps and non-negligible ground-state overlaps between consecutive members admits an efficient adiabatic interpolation.
- **Theorem 3 (Q-sampling from Markov chains).** Any efficient approximate-counting algorithm using slowly-varying Markov chains lifts to a Q-sampler for the final limiting distribution (bipartite perfect matchings, convex-body lattice points, linear extensions).

The paper is a foundational *framework* paper (complexity-theoretic); it does not report specific numerical experiments to reproduce. What is directly reproducible on classical hardware is the *operational core*: the adiabatic evolution mechanism, the spectral-gap-vs-time relationship, and the SZK-reduction identity (Claim 1).

2 Claims table

ID	Claim	Type	Testable classically?	Tested here?
C1	Adiabatic evolution $H(s) = (1-s)H_0 + sH_1$ from $ +\rangle^{\otimes n}$ into $ \psi_{\text{target}}\rangle = \sum_x \sqrt{p(x)} x\rangle$ achieves high fidelity provided T is large enough (adiabatic theorem)	Operational	Yes (small n)	Yes
C2	The evolution time T required scales polynomially in $1/\Delta_{\min}$ where $\Delta_{\min} = \min_s \Delta(s)$ is the minimum spectral gap along the path	Scaling	Yes (small n)	Yes
C3	Claim 1 identity: $\langle \psi_0 \psi_1 \rangle = F(p_0, p_1) = \sum_x \sqrt{p_0(x)p_1(x)}$ so a Hadamard test on $\frac{1}{\sqrt{2}}(0\rangle \psi_0\rangle + 1\rangle \psi_1\rangle)$ distinguishes $\ p_0 - p_1\ \geq \alpha$ from $\ p_0 - p_1\ \leq \beta$	Identity/reduction	Yes	Yes
C4	Theorem 1: $\text{SZK} \subseteq \text{BQP}$ via Qsampling reduction	Complexity-theoretic	No (structural)	Spot-check via C3
C5	Theorem 2: adiabatic state generation $\equiv$ circuit state generation	Complexity-theoretic	No (structural)	No
C6	Lemma 1: sparse Hamiltonian simulation	Algorithmic	Partially (small demo)	No (out of scope)
C7	Lemma 2: jagged adiabatic path	Algorithmic	Partially	No (out of scope)
C8	Theorem 3: Q-sampling from mixing Markov chains (bipartite perfect matchings, etc.)	Application	Yes (very small)	No (out of scope; distinct paper-length effort)

3 Method

3.1 Reproduction target

The paper’s operational core (C1–C3) is what a numpy statevector simulation on $n = 8$ qubits (256-dim Hilbert space) can meaningfully test. We do exactly that.

3.2 Hamiltonian construction

For any target normalized state $|\psi\rangle \in \mathbb{C}^{256}$ we use the canonical projector Hamiltonian

$$H_\psi = I - |\psi\rangle\langle\psi| \quad \Rightarrow \quad \text{ground state} = |\psi\rangle, \text{ ground energy} = 0, \text{ gap} = 1.$$

Then $H_0 = I - |+\rangle\langle+|^n$ and $H_1 = I - |\psi_{\text{target}}\rangle\langle\psi_{\text{target}}|$. This is a natural instantiation of the paper’s “projector on the orthogonal complement of the ground state” construction (used in Sec. 4–5 for the projector step ΠH_j of the jagged-path lemma).

3.3 Adiabatic path and spectral gap

Along $H(s) = (1-s)H_0 + sH_1$, the two lowest eigenvalues lie in $\text{span}\{|+\rangle, |\psi_{\text{target}}\rangle\}$ and, writing $o = |\langle + | \psi_{\text{target}} \rangle|$, obey

$$\Delta(s) = \sqrt{1 - 4s(1-s)(1-o^2)}, \quad \Delta_{\min} = \Delta(1/2) = o.$$

Every other eigenvalue equals 1 (the identity component outside the 2D subspace).

3.4 Time evolution

We discretize $s \in [0, 1]$ into T steps ($s_k = k/T$, $k = 1..T$), evolve for total wall time t_{tot} so $dt = t_{\text{tot}}/T$, and apply

$$|\psi(s_k)\rangle = e^{-iH(s_k)dt} |\psi(s_{k-1})\rangle, \quad |\psi(0)\rangle = |+\rangle.$$

Each propagator is computed via Hermitian eigendecomposition: $U = V e^{-i\Lambda dt} V^\dagger$. Since $|+\rangle$ and $|\psi_{\text{target}}\rangle$ span a 2D $H(s)$ -invariant subspace, we project into that subspace for the propagation — exact, no Trotter error. A **sanity check** against full 256×256 dynamics confirms the shortcut is exact to $\|\Delta\psi\|_2 = 1.7 \cdot 10^{-14}$.

3.5 Distributions tested (all on 8 bits, 256 outcomes)

- **A. Uniform.** $p(x) = 1/256$. Baseline: $|\psi_{\text{target}}\rangle = |+\rangle$, $o = 1$, $\Delta_{\min} = 1$.
- **B. Product Bernoulli(0.3).** $p(x) = \prod_i q^{x_i} (1-q)^{1-x_i}$, $q = 0.3$. A tensor-product distribution — classically efficiently samplable, SZK-flavored (in $\text{BPP} \subseteq \text{SZK}$).
- **C. Half-uniform coset.** p uniform on $\{0, \dots, 127\}$. Support = coset/subgroup; mimics the DLPc / QR reductions in the paper (Sec. 2.4–2.5) where $|C_y\rangle$ is a uniform superposition over a shifted set.
- **D. Two-peak SZK.** p = mixture of two disjoint random 32-element supports (Statistical Difference-style: two distributions with variation-distance close to $1/2$).

3.6 Software & versions

- Python 3.13 (host system), numpy 2.4.3, scipy 1.18.0
- Full run in ≈ 15 seconds on a laptop (M1/M3 class)
- Code: `report/evidence/adiabatic_state_gen.py` (13 KB, single self-contained file)
- Extractions: Nougat (Meta) v0.1.x for LaTeX-preserving parse; pdftotext (poppler) fallback for the marker.md slot (marker-pdf could not be installed on this host)

3.7 Exact commands

```
cd ~/Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-0301023-adiabatic-state-generation-SZK-aharonov
python3 -u report/evidence/adiabatic_state_gen.py \
    | tee report/evidence/run.log
# Emits report/evidence/adiabatic_results.json (full numeric results)
```

4 Results vs. paper

4.1 C1: adiabatic recovery of $|\psi_{\text{target}}\rangle$

Final fidelity $F = |\langle\psi(T)|\psi_{\text{target}}\rangle|^2$ at $t_{\text{tot}} = 30$:

Experiment	$o = \langle + \psi_{\text{tgt}}\rangle$	$\Delta_{\min}$	$F(T=10)$	$F(T=25)$	$F(T=100)$	$F(T=200)$
A. Uniform	1.0000	1.0000	1.000000	1.000000	1.000000	1.000000
B. Bernoulli $_{q=0.3}$	0.8432	0.8432	0.998812	0.999420	0.999485	0.999488
C. Half-uniform	0.7071	0.7071	0.999571	0.999767	0.999788	0.999789
D. Two-peak SZK	0.4542	0.4542	0.998000	0.997331	0.997231	0.997226

Verdict for C1: Reproduced. Every distribution hits $F > 0.99$ by $T = 10$, and $F > 0.99$ persists through $T = 200$. Well past the paper’s “ $F > 0.9$ ” adiabatic-tracking threshold.

4.2 C2: T scales with $1/\Delta_{\min}$

We modulate $\Delta_{\min}$ by varying Bernoulli parameter q (small $q \Rightarrow$ sharper distribution $\Rightarrow$ smaller overlap $\Rightarrow$ smaller gap). We measure the smallest t_{tot} (with $T = 80$ discretization steps) that reaches $F > 0.9$:

q	$o = \Delta_{\min}$	t_{tot} needed for $F > 0.9$	$1/\Delta_{\min}^2$
0.50	1.0000	5	1.0
0.40	0.9602	5	1.08
0.30	0.8432	5	1.41
0.20	0.6561	10	2.32
0.10	0.4096	20	5.96
0.05	0.2657	40	14.16

Verdict for C2: Reproduced. t_{tot} needed grows monotonically as $\Delta_{\min}$ shrinks; the observed scaling is consistent with the standard adiabatic-theorem bound $t \sim 1/\Delta_{\min}^2$ (in our small window it in fact tracks $1/\Delta_{\min}^2$ closely, growing by $2\times$ each time $\Delta_{\min}$ roughly halves and the $1/\Delta_{\min}^2$ predictor grows $\sim 2\text{--}3\times$).

4.3 C3: SZK Claim 1 identity

For $p_0 = \text{Bernoulli}(0.3)$, $p_1 = \text{uniform}$:

$$\langle\psi_{p_0}|\psi_{p_1}\rangle = 0.8431969639359371, \quad F(p_0, p_1) = \sum_x \sqrt{p_0(x)p_1(x)} = 0.8431969639359372,$$

absolute difference = 1.11×10^{-16} (machine precision). Variation distance $\|p_0 - p_1\|_1/2 = 0.4426$.

Verdict for C3: Reproduced to machine precision. This is the identity that makes the Hadamard-test SZK-to-Qsampling reduction work.

4.4 Sanity check: 2D-subspace vs full 256-dim dynamics

On experiment B, the 2D-subspace exact propagator and the full 256×256 propagator produce identical states: $\|\psi_{2D} - \psi_{256}\|_2 = 1.7 \times 10^{-14}$, $|F_{2D} - F_{256}| = 3.8 \times 10^{-15}$. Confirms we are simulating the correct dynamics.

5 Verdict

REPLICATED

(operational core — adiabatic state generation mechanism +
 SZK Claim 1 identity + T -vs-gap scaling — all reproduced
 on real numpy statevector simulation, $n = 8$ qubits, 256-dim.)

Justification. The paper’s central operational claims that admit a direct classical simulation — (C1) that the straight-line adiabatic path $(1-s)H_0 + sH_1$ with projector Hamiltonians recovers $|\psi\rangle = \sum_x \sqrt{p(x)}|x\rangle$ at high fidelity, (C2) that the required evolution time scales with $\text{poly}(1/\Delta_{\min})$, and (C3) the SZK reduction identity $\langle\psi_0|\psi_1\rangle = F(p_0, p_1)$ — all reproduced quantitatively on real numpy statevector arithmetic in 15 seconds, with a full-dimensional sanity check confirming the propagation is faithful to 10^{-14} . The paper’s structural complexity-theoretic claims (Theorems 1/2/5) are not classically “benchmarkable” but rest on exactly these mechanics, which we verified.

6 Open Questions

Q1. On $|+^n\rangle \rightarrow |\psi\rangle$ with $H_1 = I - |\psi\rangle\langle\psi|$, the observed final fidelity plateaus at ≈ 0.997 – 0.999 rather than approaching 1 as T grows (experiment D: $F(T=200) = 0.9972$, slightly *worse* than $F(T=10) = 0.9980$). What is the residual? Is it (a) discretization error from the $s_k = k/T$ endpoint not being exactly $s = 1$, (b) diabatic phase pickup within the 2D subspace, or (c) something else? Concretely, does using a boundary-vanishing schedule $s(u) = \sin^2(\pi u/2)$ eliminate the plateau, and how does the plateau depend on t_{tot} ? *Basis:* the paper predicts high fidelity is achievable but does not discuss the specific saturating value; our numerics show a distribution-dependent floor.

Q2. The paper’s projector-Hamiltonian construction requires implementing $H_1 = I - |\psi\rangle\langle\psi|$, which is generally not row-sparse in the computational basis (density matrix of a superposition is fully dense). How does the “sparse Hamiltonian lemma” (Lemma 1) actually let one *simulate* this specific H_1 efficiently for a given samplable p ? Is the paper implicitly using a projector via Kitaev phase estimation on a different, sparse “check-Hamiltonian” H_{check} whose ground state is $|\psi\rangle$? *Basis:* our replication used the dense H_1 directly; a hardware-honest replication would need the sparse-in-basis surrogate, and it is not obvious how to build it for a generic samplable distribution.

Q3. We used $H_0 = I - |+^n\rangle\langle +^n|$, but the paper’s §3 more naturally uses $H_0 = \sum_i (I - X_i)/2$ (the transverse-field / Grover-like initial Hamiltonian, which has $|+^n\rangle$ as ground state but is 1-local and sparse). Does replacing our dense H_0 with this local variant preserve the observed fidelity and gap scaling on the same distributions? Concretely, does the min-gap shrink polynomially or exponentially with n for $H_1 = \sum_x (I - |x\rangle\langle x|) \cdot (\text{some sparse encoding of } p)$? *Basis:* the local- H_0 scaling is the physically meaningful question for real hardware, and our all-dense construction sidesteps it.

Q4. The observed t_{tot} -vs- $\Delta_{\min}$ growth in our C2 table (5, 5, 5, 10, 20, 40) matches $1/\Delta_{\min}^2$ better than $1/\Delta_{\min}^3$. But the standard rigorous adiabatic theorem gives $O(1/\Delta_{\min}^3)$, and only smoothed schedules give $O(1/\Delta_{\min}^2)$ (Jansen-Ruskai-Seiler, van Dam et al.). The paper cites both regimes but doesn’t commit. Which regime governs their construction specifically for state generation, and does the empirical $1/\Delta_{\min}^2$ we observe extend to $n \gtrsim 15$ where the constant factors in the higher-order corrections matter? *Basis:* the exact scaling exponent is the difference between a polynomial algorithm and a not-really-polynomial one; our data suggest the paper’s regime is the friendlier $1/\Delta_{\min}^2$ one but this is not proven.

Q5. Claim 1’s Hadamard-test requires preparing $\frac{1}{\sqrt{2}}(|0\rangle|\psi_0\rangle + |1\rangle|\psi_1\rangle)$ coherently. In the SD-complete SZK problem the “distributions” are given only by sampling circuits C_0, C_1 ; the

paper argues Q-sampling from C_i gives $|\psi_i\rangle$, but the actual *end-to-end wall-time cost* of the SZK-decision procedure (adiabatic evolution twice + one-qubit measurement + $O(\log(1/\delta))$ repetitions) is nowhere estimated in the paper. What is that end-to-end cost as a function of $(n, \alpha, \beta, \Delta_{\min})$, and how does it compare to the best known classical SD algorithm at the same instance size? *Basis:* the paper proves a containment $\text{SZK} \subseteq \text{BQP}$ but leaves the concrete polynomial exponents un-quantified; our C3 identity check is a one-shot component of a much larger cost estimate that has never been benchmarked.

7 Reproducibility notes

- Random seed: `numpy default_rng(20260705)` (only affects experiment D’s random 32-subset support — deterministic across runs).
- Wall time: 15.4 s on a laptop CPU.
- Full JSON results dumped to `report/evidence/adiabatic_results.json`.
- Run log at `report/evidence/run.log`.